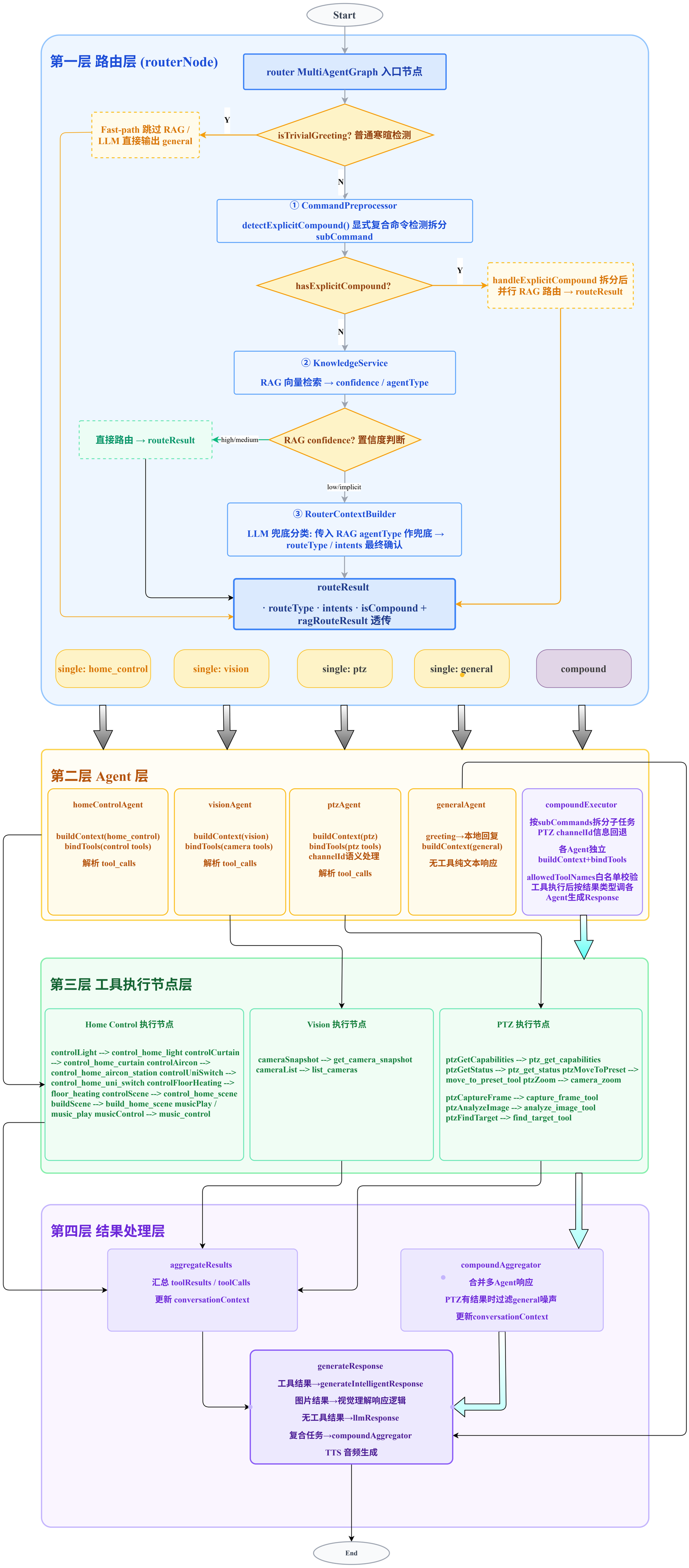
Start
第一层 路由层 (routerNode)
router MultiAgentGraph 入口节点
isTrivialGreeting? 普通寒暄检测
Y
Fast-path 跳过 RAG / LLM 直接输出 general
N
① CommandPreprocessor
detectExplicitCompound() 显式复合命令检测拆分 subCommand
hasExplicitCompound?
Y
handleExplicitCompound 拆分后并行 RAG 路由 → routeResult
N
② KnowledgeService
RAG 向量检索 → confidence / agentType
RAG confidence? 置信度判断
high/medium
直接路由 → routeResult
low/implicit
③ RouterContextBuilder
LLM 兜底分类: 传入 RAG agentType 作兜底 → routeType / intents 最终确认
routeResult
· routeType · intents · isCompound + ragRouteResult 透传
single: home_control
single: vision
single: ptz
single: general
compound
第二层 Agent 层
homeControlAgent
buildContext(home_control)
bindTools(control tools)
解析 tool_calls
visionAgent
buildContext(vision)
bindTools(camera tools)
解析 tool_calls
ptzAgent
buildContext(ptz)
bindTools(ptz tools)
channelId语义处理
解析 tool_calls
generalAgent
greeting→本地回复
buildContext(general)
无工具纯文本响应
compoundExecutor
按subCommands拆分子任务
PTZ channelId信息回退
各Agent独立
buildContext+bindTools
allowedToolNames白名单校验
工具执行后按结果类型调各
Agent生成Response
第三层 工具执行节点层
Home Control 执行节点
controlLight --> control_home_light controlCurtain --> control_home_curtain controlAircon --> control_home_aircon_station controlUniSwitch --> control_home_uni_switch controlFloorHeating --> floor_heating controlScene --> control_home_scene buildScene --> build_home_scene musicPlay / music_play musicControl --> music_control
Vision 执行节点
cameraSnapshot --> get_camera_snapshot
cameraList --> list_cameras
PTZ 执行节点
ptzGetCapabilities --> ptz_get_capabilities
ptzGetStatus --> ptz_get_status ptzMoveToPreset --> move_to_preset_tool ptzZoom --> camera_zoom
ptzCaptureFrame --> capture_frame_tool
ptzAnalyzeImage --> analyze_image_tool
ptzFindTarget --> find_target_tool
第四层 结果处理层
aggregateResults
汇总 toolResults / toolCalls
更新 conversationContext
compoundAggregator
合并多Agent响应
PTZ有结果时过滤general噪声
更新conversationContext
generateResponse
工具结果→generateIntelligentResponse
图片结果→视觉理解响应逻辑
无工具结果→llmResponse
复合任务→compoundAggregator
TTS 音频生成
End